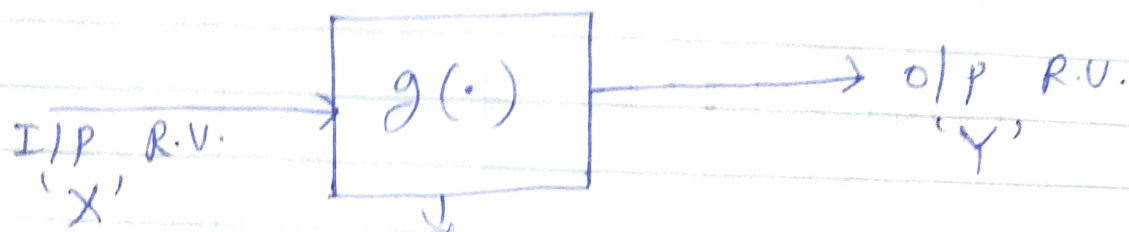


Transformation of Random Variables:-



Let $f_X(x) \rightarrow$ probability density function of I/P R.V. 'X'

$f_Y(y) \rightarrow$ Probability density function of ~~R.V.~~ O/P R.V.

$\Rightarrow$ From above transformation, we can write.

$$\boxed{Y = g(X)} \quad \text{or} \quad X = g^{-1}(Y)$$

one to one mapping b/w X & Y

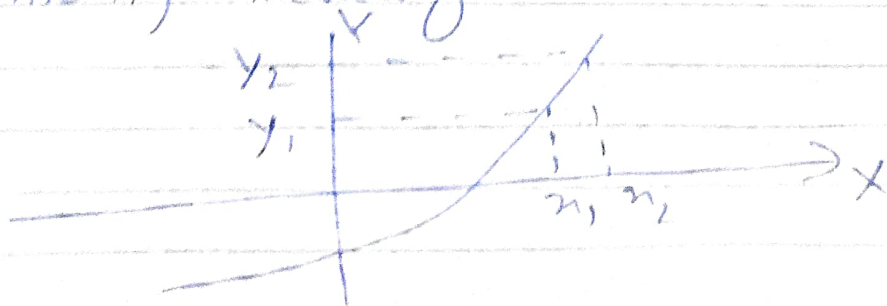
Case 1: If transformation are monotonically increasing.

$$\text{i.e. for } \begin{matrix} x_1 & , & y_1 = g(x_1) \\ x_2 & , & y_2 = g(x_2) \end{matrix}$$

$$\text{if } x_2 > x_1 \\ \Rightarrow g(x_2) > g(x_1)$$

$$\Rightarrow \boxed{y_2 > y_1}$$

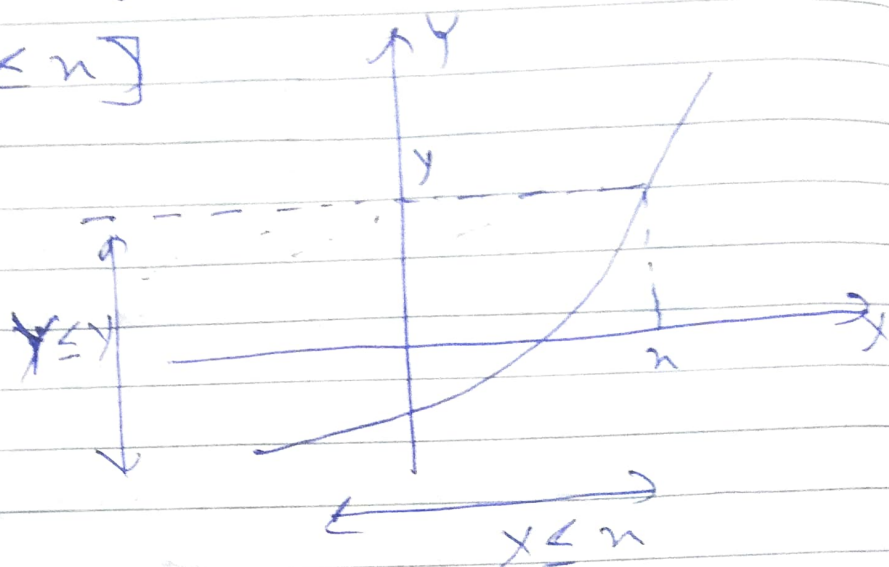
then transformation is said to be monotonically increasing.



$\Rightarrow$ CDF of R.V. Y is given as:

$$F_Y(y) = P[Y \leq y]$$

$$F_Y(y) = P[X \leq n]$$



$$\Rightarrow F_Y(y) = \int_{-\infty}^n f_X(u) du$$

$$\Rightarrow \boxed{F_Y(y) = F_X(n)}$$

For monotonically increasing function.

differentiating both side w.r.to 'y'

$$\Rightarrow \frac{d}{dy} F_Y(y) = \frac{d}{dy} F_X(n) \cdot \frac{dn}{dy}$$

$$\Rightarrow f_Y(y) = \frac{dn}{dy} \cdot f_X(n)$$

$$\Rightarrow \boxed{f_Y(y) = \frac{f_X(n)}{\frac{dy}{dn}}}$$

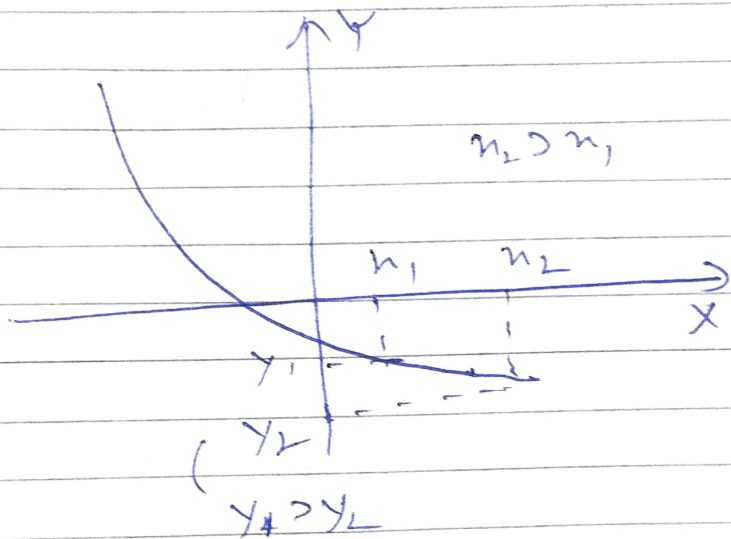
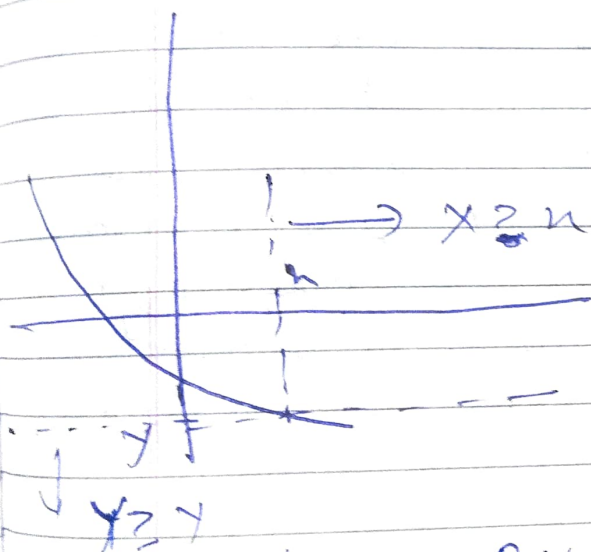
Case 2: If transformation is monotonically decreasing:

$$Y = g(X)$$

$$\text{for } x_1 \rightarrow y_1 = g(x_1)$$

$$x_2 \rightarrow y_2 = g(x_2)$$

if $x_2 > x_1 \Rightarrow y_2 < y_1$ then transformation is monotonically decreasing.



therefore R.V. 'Y' is given as.

$$F_Y(y) = P[Y \leq y]$$

$$= P[X \geq x]$$

$$= 1 - P(X \leq x)$$

$$= \boxed{F_Y(y) = 1 - F_X(x)}$$

Differentiating both side w.r.to y

$$\Rightarrow \frac{d}{dy} F_Y(y) = 0 - \frac{d F_X(u)}{du} \cdot \frac{du}{dy}$$

$$\Rightarrow f_Y(y) = - \frac{du}{dy} \cdot f_X(u)$$

$$\Rightarrow f_Y(y) = - \frac{f_X(u)}{dy/du}$$

In general:

$$f_Y(y) = \frac{f_X(u)}{|\text{slope}|}$$

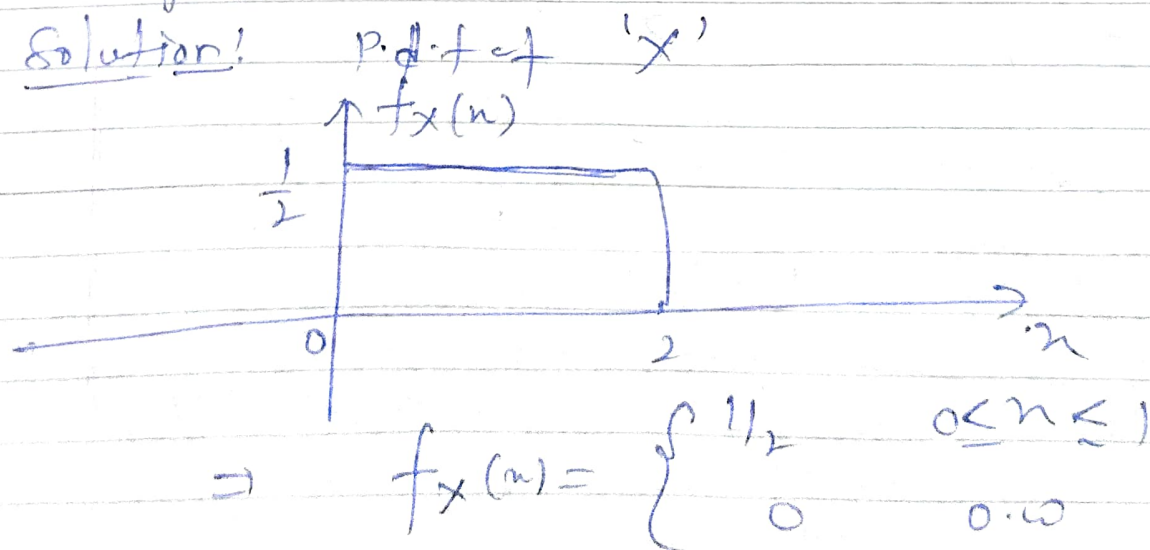
Slope = +ve
for monotonically
increasing
Slope = -ve
for monotonically
decreasing.

Ex: (1) If 'X' is uniformly distributed R.V. from 0 to 2 and 'Y' is another R.V. such that

$$Y = 2X + 1$$

then find the pdf of random variable Y. also calculate mean and variance of R.V.

Solution:



⇒ Since $Y = 2X + 1$

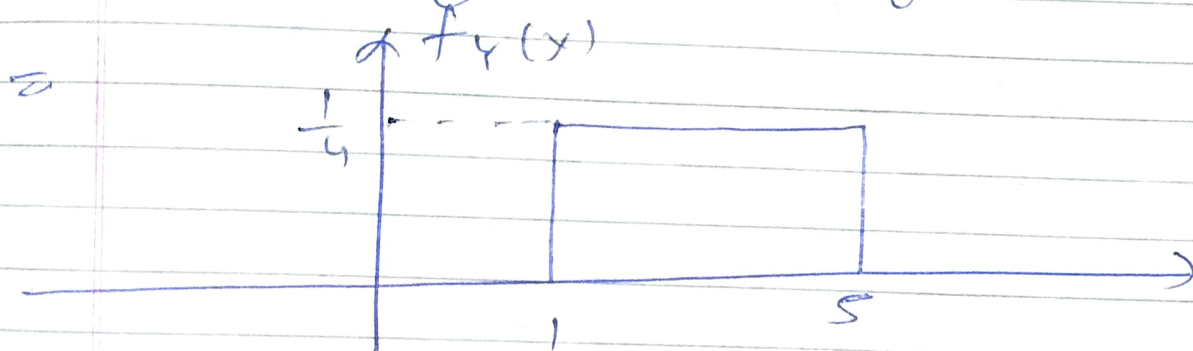
Slope = +2 (Monotonically increasing R.V.)

⇒ we know that

$$f_Y(y) = \frac{f_X(x)}{|\text{slope}|} = \frac{f_X(x)}{2}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{2} & 0 \leq y \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{4} & 1 \leq y \leq 5 \\ 0 & \text{o.w.} \end{cases}$$



$$\Rightarrow \text{mean} = \frac{a+b}{2} = \frac{1+5}{2} = \frac{6}{2} = 3$$

$$\Rightarrow \text{Variance} = \frac{(a-b)^2}{12} = \frac{(1-5)^2}{12} = \frac{16}{12} = \frac{4}{3}$$

Ex 2: Let 'X' is R.V. having Gaussian pdf with $\mu = 2$, $\sigma^2 = 4$, Y is another R.V. defined as

$$Y = -2X + 3$$

Find pdf of R.V. Y.

Solution:- we know that

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\Rightarrow f_x(x) = \frac{1}{\sqrt{8\pi}} e^{-\frac{(x-2)^2}{8}}$$

Since $y = -2x + 3$

$$\Rightarrow f_y(y) = \frac{f_x(x)}{|-2|} = \frac{f_x(x)}{2}$$

$$\Rightarrow f_y(y) = \frac{1}{2\sqrt{8\pi}} e^{-\frac{(x-2)^2}{8}}$$

Since ~~$y = -2x + 3$~~ $y = -2x + 3$

$$\Rightarrow -2x = y - 3$$

$$\Rightarrow x = \frac{3-y}{2}$$

$$\Rightarrow f_y(y) = \frac{1}{2\sqrt{8\pi}} e^{-\left(\frac{3-y}{2} - 2\right)^2 / 8}$$

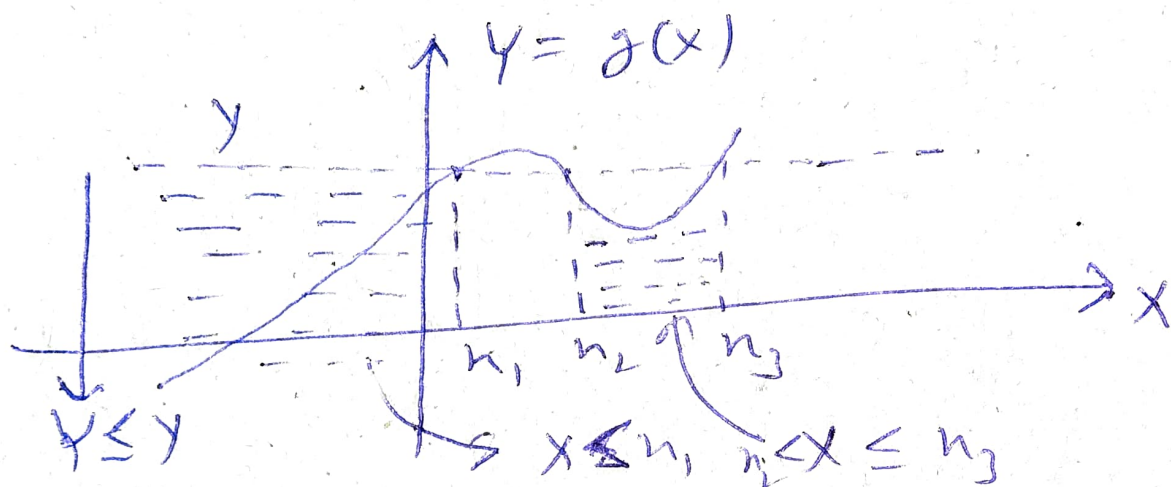
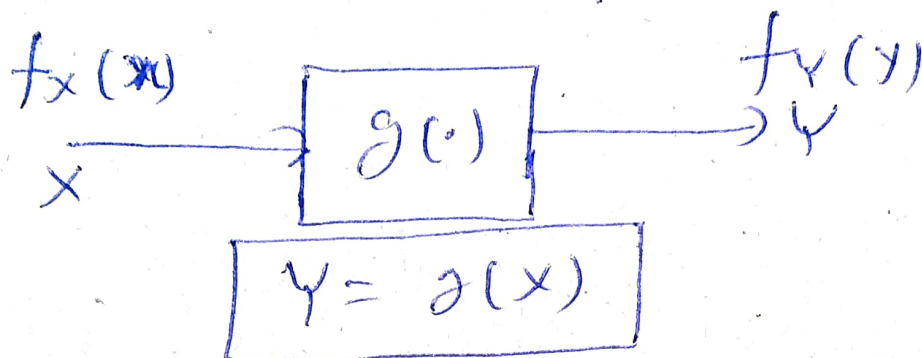
$$\Rightarrow f_y(y) = \frac{1}{\sqrt{32\pi}} e^{-\left(\frac{-y-1}{2}\right)^2 / 8}$$

$$\Rightarrow f_y(y) = \frac{1}{\sqrt{32\pi}} e^{-(y+1)^2 / 32}$$

Ans

Lecture 2 @ unit 3

(*) Non-Monotonic Transformation of Random Variables!

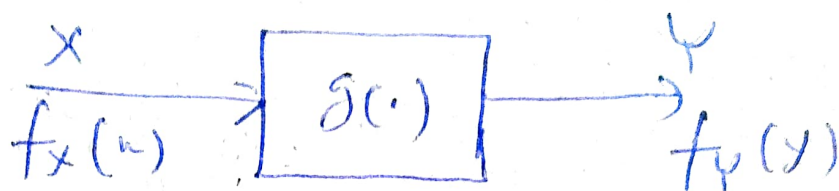


$$\Rightarrow P[Y \leq y] = P[X \leq x_1] + P[x_2 < X \leq x_3]$$

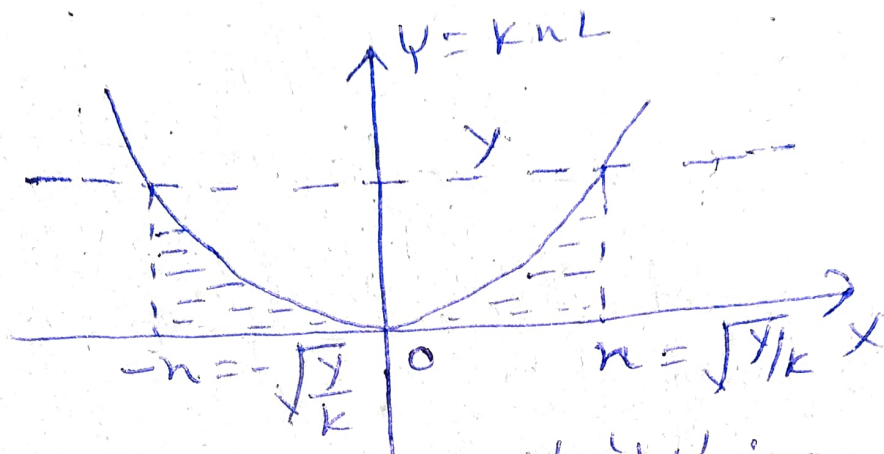
$$\Rightarrow f_Y(y) = \sum_n \frac{f_X(x_n)}{\left| \frac{dy}{dx} \right|_{x=x_n}}$$

$$\Rightarrow f_Y(y) = \frac{f_X(x_1)}{\left| \frac{dy}{dx} \right|_{x=x_1}} + \frac{f_X(x_2)}{\left| \frac{dy}{dx} \right|_{x=x_2}} + \frac{f_X(x_3)}{\left| \frac{dy}{dx} \right|_{x=x_3}}$$

Example!



$$Y = kn^2$$



CDF of Random Variable ' Y ' :-

$$F_Y(y) = P(Y \leq y)$$

$$\Rightarrow F_Y(y) = P(-n \leq x \leq n)$$

$$\Rightarrow F_Y(y) = F_X(x) - F_X(-n)$$

PDF of Random Variable ' Y ' :-

$$\Rightarrow \frac{dF_Y(y)}{dy} = \frac{dF_X(n)}{dn} \frac{dn}{dy} - \frac{dF_X(-n)}{dn} \frac{d(-n)}{dy}$$

$$\Rightarrow f_Y(y) = f_X(n) \cdot \frac{dn}{dy} + f_X(-n) \cdot \frac{dn}{dy}$$

$$\Rightarrow \boxed{f_Y(y) = \frac{[f_X(n) + f_X(-n)]}{dy/dn}}$$

Since $y = kn^2$

$$\Rightarrow \frac{dy}{dn} = 2kn$$

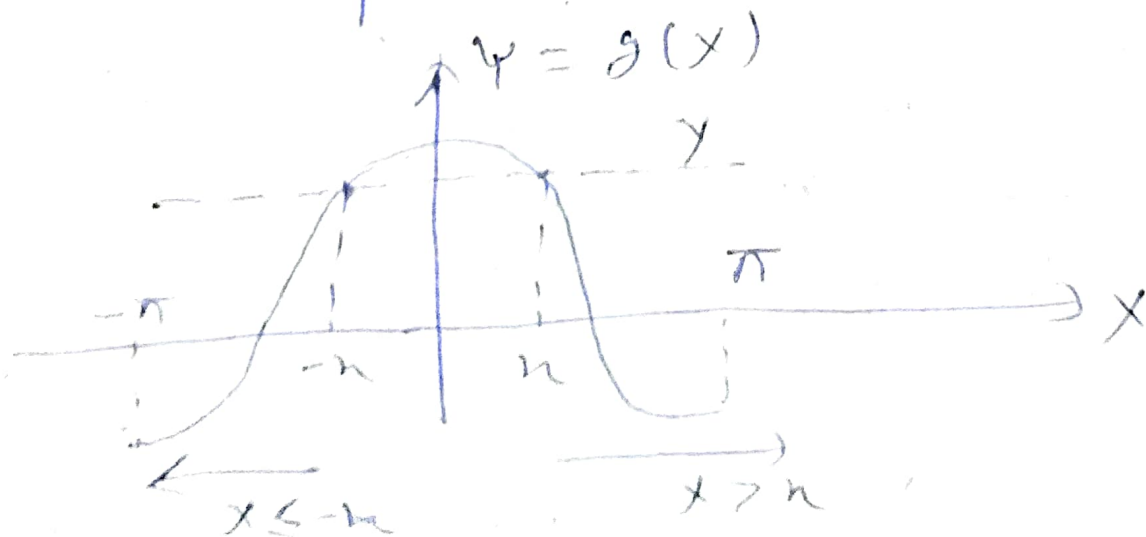
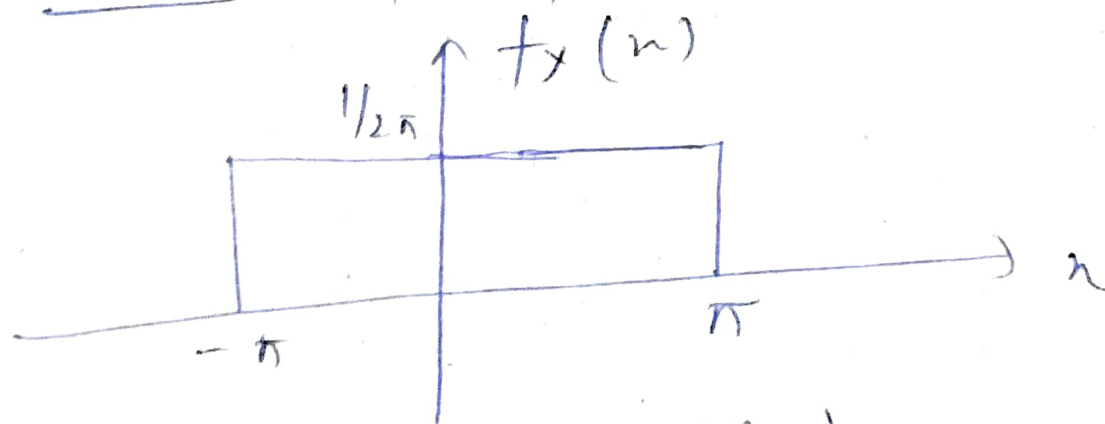
$$\Rightarrow f_Y(y) = \frac{f_X\left(\sqrt{\frac{y}{k}}\right) + f_X\left(-\sqrt{\frac{y}{k}}\right)}{2k\sqrt{y/k}}$$

→ Example 2: If 'X' is uniformly distributed R.V. from $-\pi$ to π and Y is defined as:

$$Y = \cos X$$

then find the pdf of R.V. 'Y'.

Solution: Pdf of 'X'.



CDF of R.V. 'Y' is given by:

$$F_Y(y) = P[Y \leq y]$$

$$\Rightarrow F_Y(y) = P[X \leq -n] + P[X > n]$$

$$\Rightarrow F_Y(y) = F_X(-n) + 1 - P[X \leq n]$$

$$\Rightarrow F_Y(y) = F_X(-n) + 1 - F_X(n)$$

$\Rightarrow$ Pdf of R.V. 'Y' :-

$$\Rightarrow \frac{dF_Y(y)}{dy} = \frac{dF_X(-n)}{dn} \cdot \frac{d(-n)}{dy} - \frac{dF_X(n)}{dn} \cdot \frac{dn}{dy}$$

$$\Rightarrow \frac{dF_Y(y)}{dy} = -f_X(-n) \cdot \frac{d(+n)}{dy} - f_X(n) \cdot \frac{dn}{dy}$$

$$\Rightarrow \frac{dF_Y(y)}{dy} = \frac{-[f_X(n) + f_X(-n)]}{dy/dn}$$

Since $y = \cos n$

$$\Rightarrow \frac{dy}{dn} = -\sin n$$

$$\Rightarrow f_Y(y) = \frac{-2f_X(n)}{-\sin n} = \frac{2f_X(n)}{\sin n}$$

$$\Rightarrow f_Y(y) = \frac{2}{\sin(\cos^{-1}y)} \begin{cases} \frac{1}{2\pi} & -\pi \leq y \leq \pi \\ 0 & \text{o.w} \end{cases}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{\pi \sin(\cos^{-1}y)} & -1 \leq y \leq 1 \\ 0 & \text{o.w} \end{cases}$$

P₃

Example (3):- If 'X' is uniformly distributed R.V. from -1 to 1 and Y is defined as

$$Y = X^2$$

then find the Pdf of R.V. Y.

Solution: Ans:

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}} & 0 \leq y \leq 1 \\ 0 & \text{o.w} \end{cases}$$

(*) Chebyshev's Inequality:-

If 'X' is a random variable with mean 'μ' and Variance σ_n^2 (σ), then for any positive number 'ε'

$$P[|X - \mu| \geq \epsilon \sigma_n] \leq \frac{1}{\epsilon^2}$$

or

$$P[|X - \mu| < \epsilon \sigma_n] \geq 1 - \frac{1}{\epsilon^2}$$

Proof: Let 'X' is a continuous R.V.

$$\sigma_n^2 = E[(X - \mu)^2] = E[(X - E(X))^2]$$

$$\Rightarrow \sigma_n^2 = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f_X(x) dx$$

$$\Rightarrow \sigma_n^2 = \int_{-\infty}^{\mu - \epsilon \sigma_n} (x - \mu)^2 f_X(x) dx + \int_{\mu - \epsilon \sigma_n}^{\mu + \epsilon \sigma_n} (x - \mu)^2 f_X(x) dx + \int_{\mu + \epsilon \sigma_n}^{\infty} (x - \mu)^2 f_X(x) dx$$



$$\Rightarrow \sigma_n^2 \geq \int_{-\infty}^{\mu - \epsilon \sigma_n} (n - \mu)^2 f_X(n) dn + \int_{\mu + \epsilon \sigma_n}^{\infty} (n - \mu)^2 f_X(n) dn \quad \text{--- (1)}$$

Since $n \leq \mu - \epsilon \sigma_n$ and $n \geq \mu + \epsilon \sigma_n$

$\Downarrow$

$$\cancel{\mu - \epsilon \sigma_n} \geq \mu$$

$$|(n - \mu)| \geq \epsilon \sigma_n$$

$$\Rightarrow (n - \mu)^2 \geq \epsilon^2 \sigma_n^2$$

from equation (1)

$$\sigma_n^2 \geq \int_{-\infty}^{\mu - \epsilon \sigma_n} \epsilon^2 \sigma_n^2 f_X(n) dn + \int_{\mu + \epsilon \sigma_n}^{\infty} \epsilon^2 \sigma_n^2 f_X(n) dn$$

$$\Rightarrow \sigma_n^2 \geq \epsilon^2 \sigma_n^2 \left[\int_{-\infty}^{\mu - \epsilon \sigma_n} f_X(n) dn + \int_{\mu + \epsilon \sigma_n}^{\infty} f_X(n) dn \right]$$

$$\Rightarrow \sigma_n^2 \geq \epsilon^2 \sigma_n^2 \left[P(X \leq \mu - \epsilon \sigma_n) + P(X \geq \mu + \epsilon \sigma_n) \right]$$

$$\Rightarrow \sigma_n^2 \geq \epsilon^2 \sigma_n^2 \left[P(|X - \mu| \geq \epsilon \sigma_n) \right]$$

$$\Rightarrow 1 \geq \epsilon^2 P(|X - \mu| \geq \epsilon \sigma_n)$$

$$\Rightarrow \boxed{P[|X - \mu| \geq \epsilon \sigma_n] \leq \frac{1}{\epsilon^2}} \quad \text{H.P.}$$

Since:

$$P(|X - \mu| \geq \epsilon \sigma_n) + P(|X - \mu| < \epsilon \sigma_n) = 1$$

$$\Rightarrow P(|X - \mu| < \epsilon \sigma_n) = 1 - P(|X - \mu| \geq \epsilon \sigma_n)$$

$$\Rightarrow \boxed{P[|X - \mu| < \epsilon \sigma_n] \geq 1 - \frac{1}{\epsilon^2}} \quad \text{H.P.}$$

Second form of Chebyshev's inequality:-

$$\text{let } \epsilon \sigma_n = k > 0$$

$$\Rightarrow \epsilon = k / \sigma_n$$

$$\Rightarrow P[|X - \mu| \geq k] \leq \frac{1}{k^2 / \sigma_n^2}$$

$$\Rightarrow \boxed{P[|X - \mu| \geq k] \leq \frac{\sigma_n^2}{k^2}}$$

Similarly:

$$\boxed{P[|X - \mu| < k] \geq 1 - \frac{\sigma_n^2}{k^2}}$$

Lecture 5 Unit III

(*) Random variables with non-negative values: - (Markov-inequality)

$\Rightarrow$ consider a random variable 'X' having probability density function $f_X(u) = 0$ for $u < 0$ then 'X' is said to take on only non-negative values. For such a random variable, the following inequality applies: -

$$P[X \geq \epsilon] \leq \frac{E[X]}{\epsilon} \quad \epsilon > 0$$

$\Rightarrow$ In contrast to the Chebyshev inequality, this inequality involves only the mean of X

Proof: we know that

$$E[X] = \int_{-\infty}^{\infty} u f_X(u) du$$

$$\Rightarrow E[X] = \int_0^{\infty} u f_X(u) du$$

$$\Rightarrow E[X] = \int_0^{\epsilon} u f_X(u) du + \int_{\epsilon}^{\infty} u f_X(u) du$$

$$\Rightarrow E[X] \geq \int_{\epsilon}^{\infty} u f_X(u) du$$

$$\Rightarrow E[X] \geq \int_t^{\infty} t \cdot f_X(u) du$$

$$= E[X] \geq t \int_t^{\infty} f_X(u) du$$

$$= E[X] \geq t P[X \geq t]$$

$$\Rightarrow \boxed{P[X \geq t] \leq \frac{E[X]}{t}}$$

(*) Ex: If the number of items produced in a factory during a week is a random variable with mean 500. Find the probability that this week production will be at least 1000.

Solution:

$$E[X] = 500$$

$$P[X \geq 1000] = ?$$

$$\Rightarrow P[X \geq t] \leq \frac{E[X]}{t}$$

$$\Rightarrow P[X \geq t] = \frac{500}{1000}$$

$$\boxed{P[X \geq t] = \frac{1}{2}} \quad \underline{R}$$

(*) Chernoff's bound.

$\Rightarrow$ For any $\epsilon > 0$ and $a > 0$

$$P[X \geq a] = P[e^{\epsilon X} \geq e^{\epsilon a}] \quad \text{--- (I)}$$

According to Markov-inequality, we know that

$$P[X \geq \epsilon] \leq \frac{E[X]}{\epsilon} \quad \text{--- (II)}$$

therefore from eqns (I) & (II)

$$P[X \geq a] \leq \frac{E[e^{\epsilon X}]}{e^{\epsilon a}} \quad \text{--- Chernoff's bound}$$

where $E[e^{\epsilon X}] =$ Moment generating function.

$$\Rightarrow P[X \geq a] \leq \frac{\text{Moment generating function of } X}{e^{\epsilon a}}$$

in some of books $\epsilon \rightarrow t$

$$\Rightarrow P[X \geq a] \leq \frac{E[e^{tX}]}{e^{ta}}$$

Example:- Suppose that 'X' is a random variable with mean and variance is equal to 16. Compute a lower bound on $P(0 < X < 32)$, using Chebyshev's inequality.

Solution:-

$$P(0 < X < 32)$$

$\Downarrow$

$$P(0 - 16 < X < 32 - 16)$$

$$= P(|X - 16| < 16)$$

$$= 1 - P[|X - 16| \geq 16]$$

$$= 1 - \frac{\sigma^2}{t^2}$$

$$= 1 - \frac{16}{16^2} = 1 - \frac{1}{16} = \frac{15}{16} \quad \underline{R_2}$$

Ex:-

⊗ Central limit theorem:-

Let $X_1, X_2, X_3, \dots, X_n$ are independent and identically distributed (iid) random variables, with mean μ and variance σ^2 , let take an another random variable X which is the summation of $X_1, X_2, X_3, \dots, X_n$

i.e.

$$X = X_1 + X_2 + X_3 + \dots + X_n$$

then as n increases, the distribution of X converges to the normal distribution $N(n\mu, n\sigma^2)$

Proof: Sample space (S_n)

$$S_n = X_1 + X_2 + X_3 + \dots + X_n$$

where $X_1, X_2, X_3, \dots, X_n$ are iid r.v's.

$$\Rightarrow \text{mean of } S_n = E[S_n]$$

$$\Rightarrow E[S_n] = E[X_1 + X_2 + X_3 + \dots + X_n]$$

$$\Rightarrow E[S_n] = E[X_1] + E[X_2] + E[X_3] + \dots + E[X_n]$$

$$\Rightarrow E[S_n] = \mu + \mu + \mu + \dots + \mu$$

$$\Rightarrow E[S_n] = n\mu$$

$$\text{Variance}(S_n) = \text{Variance}[X_1 + X_2 + \dots + X_n]$$

$$= \text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)$$

$$= \sigma^2 + \sigma^2 + \dots + \sigma^2$$

$$\Rightarrow \text{Var}(S_n) = n\sigma^2$$

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sqrt{n}\sigma}$$

Average of R.V.'s.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$$

$$E[\bar{X}] = \frac{E[X_1] + E[X_2] + \dots + E[X_n]}{n}$$

$$\Rightarrow E[\bar{X}] = \frac{n\mu}{n} = \mu$$

$$\Rightarrow \boxed{E[\bar{X}] = \mu}$$

$$\text{Var}(\bar{X}) = \text{Var}\left[\frac{X_1 + X_2 + X_3 + \dots + X_n}{n}\right]$$

$$= \frac{1}{n^2} [\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)]$$

$$= \frac{1}{n^2} [\sigma^2 + \sigma^2 + \dots + \sigma^2]$$

$$= \frac{n\sigma^2}{n^2}$$

$$\boxed{\text{Var}(\bar{X}) = \frac{\sigma^2}{n}}$$

$$\Rightarrow \boxed{\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)}$$

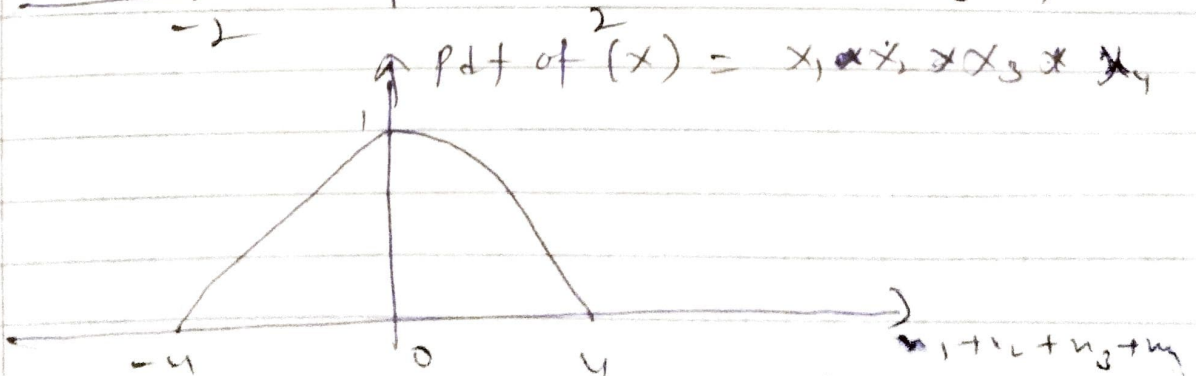
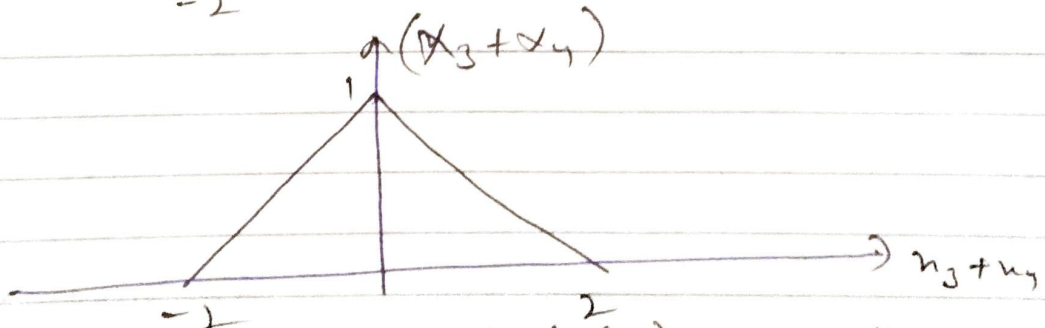
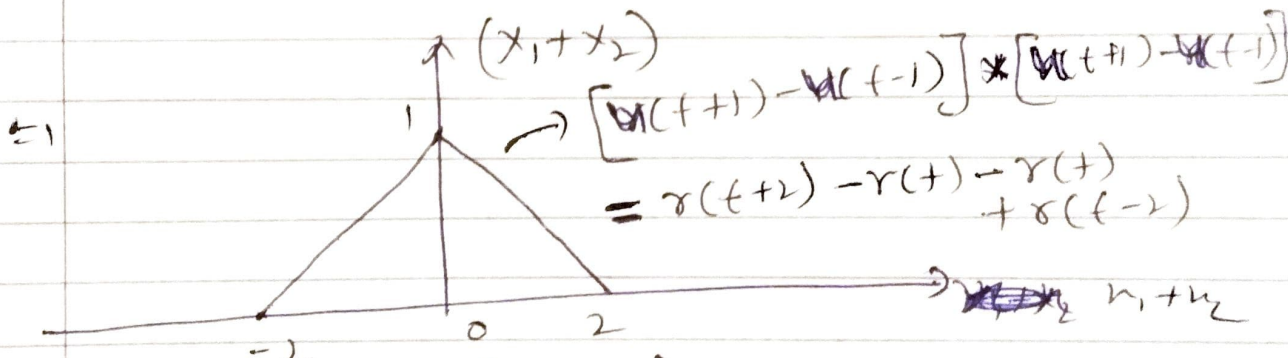
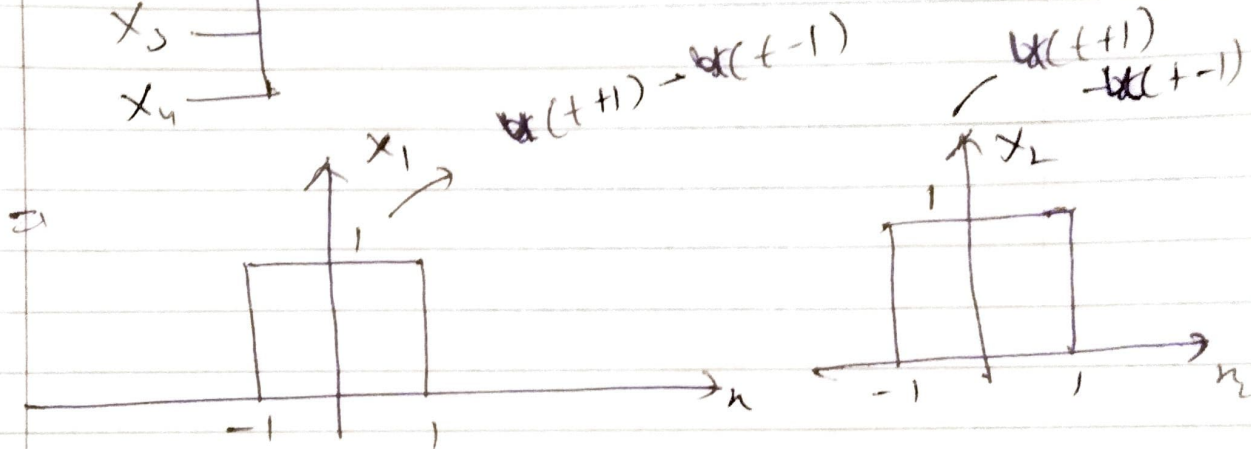
(*) Graphical proof!

$$\text{let } x = x_1 + x_2 + x_3 + x_4$$

$$\Rightarrow \text{pdf}(x) = \text{pdf}(x_1) * \text{pdf}(x_2) * \text{pdf}(x_3) * \text{pdf}(x_4)$$

let

$$\begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{matrix} \rightarrow \text{uniform}[-1, 1]$$



Lecture (6) @ unit II

(*) Multi-dimensional gaussian law:-

⇒ The one-dimensional Normal density function for a random variable 'x' is given as:

$$X \sim N(\mu, \sigma^2)$$

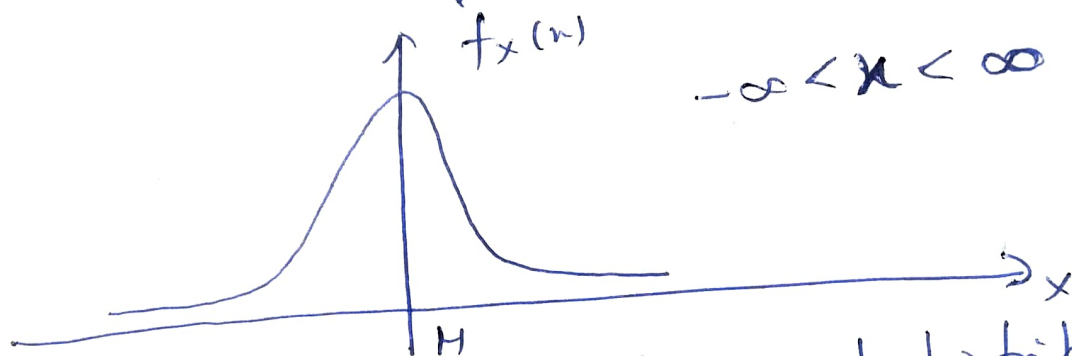
where

μ = mean of 'x'

σ^2 = Variance of 'x'

⇒ Pdf of 'x' is given as:

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$



⇒ For multi-dimensional Normal distribution:-

$$X \rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_p \end{bmatrix}_{p \times 1}$$

$$\sigma^2 = \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \dots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \dots & \sigma_{pp} \end{bmatrix}_{p \times p}$$

$$\mu \rightarrow \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \vdots \\ \mu_p \end{bmatrix}_{p \times 1}$$

⇒ Pdf of multi-dimensional Normal distribution is given as:

$$f_x(x) \sim N_p(\mu, \Sigma)$$

$\Rightarrow$ For bi-dimensional (Bivariate case)

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

or

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix}$$

$\Rightarrow$ If there is no co-relation b/w x_1 & x_2

then

$$f_X(x_1, x_2) = f_X(x_1) * f_X(x_2)$$

$$\Rightarrow f_X(x_1, x_2) = \frac{1}{\sqrt{2\pi\sigma_1^2}} e^{-\frac{1}{2}\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2} \times \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-\frac{1}{2}\left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2}$$

$$\Rightarrow f_X(x_1, x_2) = \frac{1}{(2\pi)^{\frac{2}{2}} (\sigma_1^2 \sigma_2^2)^{1/2}} e^{-\frac{1}{2} \left[\left(\frac{x_1 - \mu_1}{\sigma_1}\right)^2 + \left(\frac{x_2 - \mu_2}{\sigma_2}\right)^2 \right]} \quad \text{--- (A)}$$

$\Rightarrow$ if x_1 & x_2 are independent R.V. then

$$\sigma_{12} = \sigma_{21} = 0 \quad (\text{no co-relation b/w } x_1 \text{ & } x_2)$$

$$\Rightarrow \Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}$$

$$\Rightarrow |\Sigma| = \sigma_1^2 \sigma_2^2 - (\overset{0}{\sigma_{12}})^2$$

$$\Rightarrow |\Sigma| = \sigma_1^2 \sigma_2^2 \Rightarrow |\Sigma|^{1/2} = (\sigma_1^2 \sigma_2^2)^{1/2}$$

from eqn (A)

$$f_X(n_1, n_2) = \frac{1}{(2\pi)^{2/2} (\sigma_1^2 \sigma_2^2)^{1/2}} e^{-\frac{1}{2} \left[\left(\cdot \right) + \left(\cdot \right) \right]}$$

$$\Rightarrow f_X(n_1, n_2) = \frac{1}{(2\pi)^{2/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} \left[\left(\frac{n_1 - \mu_1}{\sigma_1} \right)^2 + \left(\frac{n_2 - \mu_2}{\sigma_2} \right)^2 \right]} \quad \text{--- (B)}$$

$\Rightarrow$ For 1-D Normal distribution, we know that

$$f_X(n) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \left[\frac{n - \mu}{\sigma} \right]^2} \quad \text{--- (C)}$$

exponent of above pdf is given as:

$$\text{exponent} = -\frac{1}{2} \left[\frac{n - \mu}{\sigma} \right]^2$$

$$\text{exponent} = -\frac{1}{2} (\sigma^2)^{-1} (n - \mu)^2$$

$$\text{exponent} = -\frac{1}{2} (\sigma^2)^{-1} (n - \mu) (n - \mu)^T$$

[since $A A^T = A^2$]

$$\Rightarrow \text{exponent} = -\frac{1}{2} \Sigma^{-1} (n - \mu) (n - \mu)^T$$

for multi-dimensional R.V. we can write,

$$\text{exponent} = -\frac{1}{2} \Sigma^{-1} (X - \mu)^T (X - \mu)$$

from eq. (B), pdf of multi-dimensional Gaussian ~~is~~ gaussian distribution is given by:

$$f_X(x_1, x_2, \dots, x_p) = \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2} [(X-M)^T \Sigma^{-1} (X-M)]}$$

Ex! Let random Variable 'X' which distribution is given by:

$$X \sim N_3(M, \Sigma)$$

where

$$M = \begin{bmatrix} 5 \\ 3 \\ 7 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & 2 \\ 0 & 2 & 9 \end{bmatrix}$$

Find:

$$(a) P[X_1 > 6]$$

$$(b) P[5X_2 + 4X_3 > 70]$$

Solution!

$$X_1 \sim N(M_1, \sigma_1^2) \quad (b) \text{ do your self.}$$

$$\Rightarrow X_1 \sim N(5, 4)$$

$$\text{we know that } Z = \frac{X-M}{\sigma}$$

$$\Rightarrow Z = \frac{X_1 - M_1}{\sigma_1} = \frac{6-5}{\sqrt{4}} = \frac{6-5}{2} = \frac{1}{2}$$

$$\Rightarrow Z = 0.5$$

$$\Rightarrow P(X_1 > 0.5) = P(Z > 0.5) \cong P(Z < -0.5)$$

$$\cong 0.3085 \quad (\text{From table of Normal Dist.})$$